Representation and Modeling of 3D Objects - Examples

Examples: Mass – spring model

- Mass Spring Damper Cloth Simulation 1 - YouTube,
  https://www.youtube.com/watch?v=ib1vmRDs8Vw
- Mass Spring Damper Cloth Simulation 2,
  https://www.youtube.com/watch?v=qOvb3WLAX0E
- Fast Simulation of Mass-Spring Systems Cloth Implementation,
  https://vimeo.com/252071746
- Wobbly Picture using Mass-Spring system with Box2D,
  https://www.youtube.com/watch?v=3cT6ZM3KkqY
- Fluid Simulations,
  https://www.youtube.com/watch?v=ESShqTMvZWU
- Advanced Molecular & Particle Physics Simulations,
  https://www.youtube.com/watch?v=x8Fo2slT2WA

Examples: L-Systems

- Realtime L-System Trees 02

  https://www.youtube.com/watch?v=HC8skymgu_0
- Procedurally Generated Apple Tree

  https://www.youtube.com/watch?v=ZVyi4ykVHJM
- Fractal Trees

  https://www.youtube.com/watch?v=t65lq818c78&list=PLE4y6Wd4RpZfon6KTy4TX65BznH9MpN7w
- Procedurally Generated Plant

  https://www.youtube.com/watch?v=0_bum83FuRk&index=2&list=PLE4y6Wd4RpZfon6KTy4TX65BznH9MpN7w
- 3D Real-Time Plants Modeleing with L-systems,

  https://www.youtube.com/watch?v=QJ6r8ZpQ7P8

Examples

- Free-flowing Granular Materials with Two-way Solid Coupling (video)
  - Continuum-based model
  - Efficient simulation of granular materials
  - Fully solves the internal pressure and frictional stresses in a granular material
  - Visually noticeable behaviors of granular materials
  - Freely dispersing splashes without cohesion, and a global coupling between friction and pressure
  - The full treatment of internal forces in the material also enables two-way interaction with solid bodies
  - Very small fraction of computational costs of the comparable particle-based models for granular flows

Examples

- RealFlow 2012 water simulation (video)
- Stitch Meshes for Modeling Knitted Clothing with Yarn-level Detail - SIGGRAPH 2012 (video)
- 3D Hair on Siggraph 2012 (video)
- Realflow Flooded Corridor (video)
  - Use standard fluids and emitters
  - Took 4 hours to simulate and mesh, 1 hour to render
  - Use an opaque material for the liquid
  - Expensive rendering of refractions
- Virtual Clothing

  http://www0.cs.ucl.ac.uk/research/vr/Projects/3DCentre/cloth_simulation_links.htm